TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Determination of Woven and Knitted Fabric Dimensions

Length, width, thickness and thread density measured on the flat table: the dimensional fingerprint that every later quality test on this fabric refers back to.

TE-3202 · QUALITY EVALUATION OF FABRIC AND APPAREL SESSIONAL · EXPERIMENT 1 · LEVEL 3 TERM 2

1. Aim and theory

Aim: to determine the length, usable width, thickness and thread density (ends and picks per inch for woven fabric; courses and wales per inch for knitted fabric) of conditioned samples, and to report the mean values that define the fabric under test throughout Experiments 1–7 of this course.

All measurements are made in the standard atmosphere for testing textiles — 65% relative humidity and 20 °C maintained throughout conditioning and testing. Cotton swells as it takes up moisture, so a fabric measured dry, damp or at a different tension is dimensionally a slightly different fabric. Specimens are conditioned for at least 24 h in the standard atmosphere, laid flat without tension on a smooth table, allowed to relax, and only then measured. No dimension on this sheet is taken on fabric fresh off the roll or straight from a packet.

1.1 Length and width

Length is measured along the selvedge direction with a calibrated steel rule or measuring table, the fabric laid flat and smoothed by hand without stretching. For a full piece, the length is the mean of measurements at the centre and near both selvedges. **Width** is the distance between the two selvedges measured at right angles to them, excluding any distorted selvedge fringe. Because width varies along a piece — loom tension, stentering and handling all pull it in and out — width is never a single reading: at least five positions spread over the length are measured and averaged. The width so obtained is the *usable width* on which yield, marker planning and cost per garment all depend.

1.2 Thickness

Thickness is the distance between the two faces of the fabric under a specified pressure between a circular presser foot and a flat anvil. It is not a fixed geometric property: a lofty fabric compresses. The reading therefore belongs to its pressure and foot area — the full treatment is Experiment 3 (ISO 5084).

In this survey experiment a dead-weight thickness gauge at the reference pressure gives the working value used for cross-reference.

1.3 Thread density — woven fabric

Thread density states how many threads occupy unit length in each direction:

$$\text{EPI} = \text{number of warp ends in 1 inch} \quad \cdot \quad \text{PPI} = \text{number of weft picks in 1 inch}$$

Counting is done with a counting glass (pick glass) over a marked 1-inch length, or by dissecting a measured specimen and counting every thread (the traveller method) where the weave is too fine or too distorted for direct counting. Several positions are counted because density varies across the width (warp density is usually uniform, set by the reed; weft density varies with loom let-off and take-up) and the mean is reported to the nearest whole thread.

1.4 Thread density — knitted fabric

A weft-knitted fabric has no warp or weft. Its density is counted in loops:

$$\text{CPC} = \text{courses per inch (loop rows, along the length)} \quad \cdot \quad \text{WPC} = \text{wales per inch (loop columns, across the width)}$$

$$\text{Stitch density} = \text{CPC} \times \text{WPC} \quad \text{(loops per square inch)}$$

Stitch density, together with loop length, controls knitted fabric weight, cover and extensibility far more directly than either count alone — two fabrics with the same CPC but different WPC are different fabrics. The knitted sample must be fully relaxed (dry-relaxed at minimum, and the relaxation state stated) before counting, because courses and wales per inch change markedly as a knit relaxes off the machine.

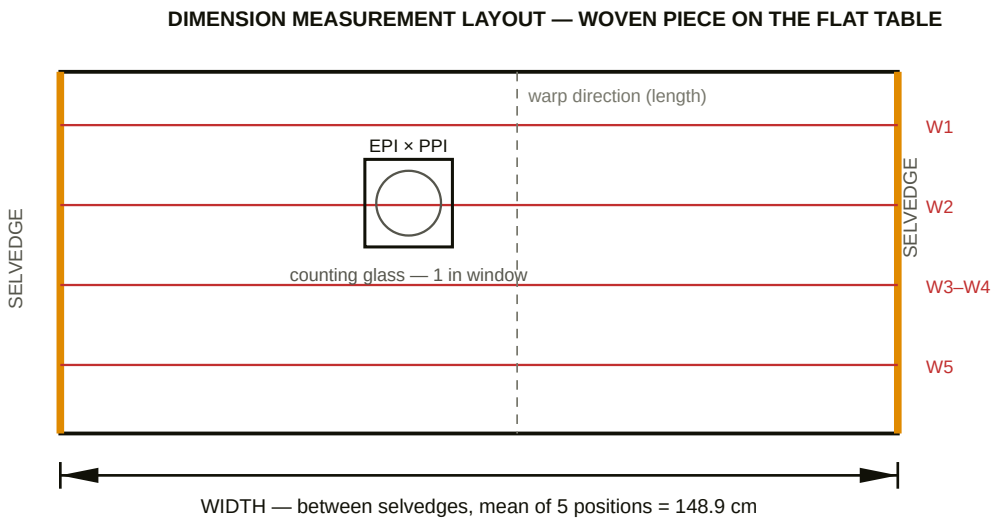


Figure 1 — A woven piece laid flat without tension: width is the selvedge-to-selvedge distance averaged over five positions (red lines), and thread density is counted with a counting glass over a marked 1-inch window.

2. Apparatus and procedure

- **Apparatus:** flat measuring table; calibrated steel rule and tape (cm/mm); counting glass with 1-inch window; dead-weight thickness gauge (presser foot on anvil, dial reading to 0.01 mm); scissors and marking pen; conditioned woven piece and knitted sample.
- **Conditioning:** condition both samples at 65% RH and 20 °C for at least 24 h. Lay them flat, smooth out creases by hand without pulling, and allow full relaxation before any reading.
- **Width (woven):** measure selvedge-to-selvedge at five positions spread over the length, at right angles to the selvedges, the rule resting lightly on the fabric. Record each reading to 0.1 cm and average.
- **Length (woven piece):** measure along the centre and near both selvedges over the full piece on the table (or in marked table lengths) and average; state the result with the width.
- **Thread density (woven):** place the counting glass flat, count the ends in 1 inch at five places across the width for EPI and the picks in 1 inch at five places along the length for PPI. Count a thread that is more than half inside the window; be consistent at both edges.
- **Thickness:** place the fabric between presser foot and anvil without wrinkling, lower the foot gently, allow the dial to settle, and read to 0.01 mm. Take ten readings at spread positions and average — full method in Experiment 3.
- **Knitted sample:** on the relaxed knit, count courses per inch along three wale lines and wales per inch along three course lines with the counting glass; average each, then multiply for stitch density.

3. Worked calculation

The woven fabric measured here is the reference plain-weave cotton fabric used in Experiments 1–7.

Width. Five readings (cm): 148.6, 149.2, 148.8, 149.0, 148.9.

$$\text{Mean width} = (148.6 + 149.2 + 148.8 + 149.0 + 148.9) / 5 = 744.5 / 5 = 148.9 \text{ cm}$$

The range is only 0.6 cm (about 0.4% of the mean) — a well-stentered piece. A range of several centimetres would warn of uneven stentering, and the *minimum* width, not the mean, would then govern marker planning.

Thread density (woven). Counting-glass counts averaged 70 ends per inch and 59 picks per inch:

$$\text{EPI} = 70 \;\cdot\; \text{PPI} = 59 \;\cdot\; \text{Thread density} = 70 \times 59$$

This 70 × 59 construction is the identity of the test fabric: Experiments 5 and 6 will show how directly it sets cover, porosity and strength.

Thickness. Ten readings at spread positions lay between 0.42 mm and 0.45 mm:

$$\text{Mean thickness} = 0.434 \text{ mm (reference pressure; see Experiment 3)}$$

Knitted sample. Relaxed counts gave CPC = 46 and WPC = 38:

$$\text{Stitch density} = \text{CPC} \times \text{WPC} = 46 \times 38 = 1{,}748 \text{ loops per square inch}$$

PROPERTY	RESULT	CROSS-REFERENCE
Mean width (woven)	148.9 cm	Used for area in Experiment 2
Thread density (woven)	70 EPI × 59 PPI	Cover and porosity in Experiment 5
Mean thickness	0.434 mm	Full pressure study in Experiment 3
Stitch density (knit)	1,748 /in²	CPC 46 × WPC 38

THE DIMENSIONAL FINGERPRINT

Width 148.9 cm · 70 × 59 threads per inch · thickness 0.434 mm (with GSM 185.5 from Experiment 2). Every later sheet quotes these same values — a strength, porosity or abrasion result is meaningless unless the fabric it belongs to is fixed this precisely.

4. Precautions

- Never measure under tension: smoothing means flattening by hand, not stretching. Even 1% stretch falsifies width, density and thickness together.
- Keep the rule square to the selvedges for width; a diagonal reading always overstates width.

- Count threads at several positions and on both faces if the weave is hard to read; a single count at the selvedge is not the fabric — selvedge density is deliberately different.
- State the relaxation state of knitted samples with the result. Courses and wales per inch of an unrelaxed knit can differ by 10% or more from the relaxed values.
- Lower the thickness-gauge foot gently and let the reading settle; dropping the foot compresses the fabric and reads thin.
- Verify the standard atmosphere (65% RH, 20 °C) before starting; if the room has drifted, re-condition rather than correct afterwards — no reliable correction factor exists.

Practice (viva preparation)

1. Why must all dimension measurements be made at 65% RH and 20 °C?
2. Five width readings are 148.6, 149.2, 148.8, 149.0 and 148.9 cm. Find the mean width and comment on the range.
3. Why is width reported as a mean of several positions rather than a single reading — and when does the minimum matter more than the mean?
4. A knitted fabric has CPC 46 and WPC 38. Find its stitch density and state what stitch density controls.
5. Why is a thickness value incomplete unless its measuring pressure is stated?
6. What is the traveller (dissection) method of counting thread density, and when is it preferred to the counting glass?
7. How do EPI and PPI in a woven fabric correspond to CPC and WPC in a knitted fabric?

Answer key

1. Cotton fibres absorb moisture and swell as relative humidity rises, changing fabric dimensions, thickness and density. The standard atmosphere of 65% RH and 20 °C fixes the moisture state so that results from different laboratories, days and fabrics are comparable. Without it, two honest measurements of the same fabric need not agree.
2. Mean = 744.5 / 5 = 148.9 cm. Range = 149.2 − 148.6 = 0.6 cm, about 0.4% — small, indicating uniform width control (good stentering). This is the reference width of the course test fabric.
3. Width varies along a piece because loom tension, take-up and finishing act unevenly. A single reading could sit on a local neck or flare and misrepresent yield and cutting calculations, so the mean of at least five spread positions is reported. For marker planning the minimum width governs, because the marker must fit at the narrowest point of the piece.
4. Stitch density = 46 × 38 = 1,748 loops per square inch. With loop length, it controls the knit's weight per unit area, cover, tightness and extensibility; it is the knitted equivalent of quoting both EPI and

PPI together.

5. Fabric is compressible: the measured gap between foot and anvil depends on how hard the foot presses. The same fabric reads about 0.52 mm at low pressure and 0.44 mm at high pressure (Experiment 3). A thickness without its pressure and foot area cannot be reproduced or compared.
6. In the traveller method a specimen of exactly measured dimensions is dissected — threads are removed one by one under a needle and counted — giving the total threads in a known width or length. It is preferred for very fine, dense, distorted or multi-layer weaves where threads cannot be resolved reliably through a counting glass.
7. EPI (ends per inch) runs along the warp, lengthwise; WPC (wales per inch) also counts lengthwise-running columns across the width — the correspondence is EPI with WPC across the width, and PPI (picks per inch, across-wise threads stacked along the length) with CPC (courses per inch stacked along the length). In short: EPI ↔ WPC measure density across the width; PPI ↔ CPC measure density along the length.
